LAB REPORT 04
Fourier Series Approximations & Line Spectra

Lab Exercise: Comprehensive Fourier Series Analysis

A periodic trapezoidal signal x(t) with period T=3 is given by:
x(t) = t for $0 \le t \le 1$
x(t) = 1 for $1 \le t \le 2$
x(t) = 3 - t for $2 \le t \le 3$

Compute and plot approximations of x(t) using 3, 11, and 51 terms for:
a. 3, 11, and 51 terms of the complex exponential Fourier series.
b. 3, 11, and 51 terms of the trigonometric Fourier series.
c. 3, 11, and 51 terms of the Fourier series in cosine with phase form.
d. Each time also plot the corresponding Fourier series coefficients.
e. Each time compute the approximation percentage.
f. Plot the line spectra.
g. Finally, plot the three ''good'' approximations in time of five periods.

MATLAB Implementation Code (exercise_1.m):

```
clear;
clc;
close all;

% 1. BASIC INFORMATION
T = 3; % Period
w0 = 2*pi/T; % Fundamental angular frequency

% Total number of complex exponential terms
termCounts = [3 11 51];
Klist = (termCounts - 1)/2;

% Time vector for one period
t = linspace(0,T,6001);

% 2. ORIGINAL SIGNAL OVER ONE PERIOD
x = zeros(size(t));
% 0 <= t < 1
index1 = (t >= 0) & (t < 1);
x(index1) = t(index1);
% 1 <= t < 2
index2 = (t >= 1) & (t < 2);
x(index2) = 1;
% 2 <= t <= 3
index3 = (t >= 2) & (t <= 3);
x(index3) = 3 - t(index3);
```

```
% Plot original signal
figure;
plot(t,x, 'k', 'LineWidth',2);
grid on;
xlabel('Time t');
ylabel('x(t)');
title('Original Signal x(t)');
xlim([0 3]);
ylim([-0.1 1.2]);

% FOURIER COEFFICIENTS
% Arrays used to save approximations
complexApprox = cell(1, length(termCounts));
trigApprox = cell(1, length(termCounts));
phaseApprox = cell(1, length(termCounts));

% Approximation percentages
PctComplex = zeros(size(termCounts));
PctTrig = zeros(size(termCounts));
PctPhase = zeros(size(termCounts));

% Energy of the original signal
signalEnergy = trapz(t,x.^2);

%% PART (a): COMPLEX EXPONENTIAL FOURIER SERIES
fprintf('\n');
fprintf(' PART (a): COMPLEX EXPONENTIAL FOURIER SERIES\n');

for q = 1:length(termCounts)
    M = termCounts(q);
    K = Klist(q);

    % Harmonic indices
    k = -K:K;

    % Calculate C_k
    Ck = zeros(size(k));
    % DC coefficient
    Ck(k==0) = 2/3;

    % Nonzero harmonics
    nz = (k~=0);
    Ck(nz) = 3*(cos(2*pi*k(nz)/3)-1) ./ (2*pi^2*k(nz).^2);

    % Construct Fourier series approximation
    E = exp(1i*(k(:)*w0)*t);
    x_complex = real(Ck * E);

    % Save result
```

```
    complexApprox{q} = x_complex;

    % Approximation percentage
    errorEnergy = trapz(t, (x - x_complex).^2);
    PctComplex(q) = 100*(1 - errorEnergy/signalEnergy);

    fprintf('%2d terms: %.6f %% approximation\n', M, PctComplex(q));

    % Plot approximation and coefficients
    figure('Name', sprintf('Complex Fourier Series %d Terms', M));

    % Approximation
    subplot(3,1,1);
    plot(t,x, 'k', 'LineWidth', 2);
    hold on;
    plot(t,x_complex, 'r--', 'LineWidth',1.5);
    grid on;
    xlabel('Time t');
    ylabel('Amplitude');
    title(sprintf('Complex Exponential FS - %d Terms, Approximation = %.6f%%', M,
PctComplex(q)));
    legend('Original x(t)', 'Fourier approximation');
    xlim([0 T]);
    ylim([-0.1 1.2]);

    % Magnitude of C_k
    subplot(3,1,2);
    stem(k, abs(Ck), 'filled');
    grid on;
    xlabel('Harmonic index k');
    ylabel('|C_k|');
    title('Magnitude of Complex Fourier Coefficients');

    % Phase of C_k
    phaseC = angle(Ck);
    % Phase of zero coefficients is undefined, so set it to zero for plotting
    phaseC(abs(Ck)<1e-12) = 0;
    subplot(3,1,3);
    stem(k,phaseC, 'filled');
    grid on;
    xlabel('Harmonic index k');
    ylabel('\angle C_k (rad)');
    title('Phase of Complex Fourier Coefficients');
end

%% PART (b): TRIGONOMETRIC FOURIER SERIES
fprintf('\n');
fprintf(' PART (b): TRIGONOMETRIC FOURIER SERIES\n');
```

```
for q = 1:length(termCounts)
    M = termCounts(q);
    K = Klist(q);

    % Positive harmonics
    n = 1:K;

    % Trigonometric Fourier coefficients
    a0 = 4/3;
    an = 3*(cos(2*pi*n/3)-1) ./ (pi^2*n.^2);

    % Because the periodic signal is even
    bn = zeros(size(n));

    % Construct approximation
    theta = (n(:)*w0)*t;
    x_trig = a0/2 + an*cos(theta) + bn*sin(theta);

    trigApprox{q} = x_trig;

    % Approximation percentage
    errorEnergy = trapz(t, (x - x_trig).^2);
    PctTrig(q) = 100*(1 - errorEnergy/signalEnergy);

    fprintf('%2d terms: %.6f %% approximation\n', M, PctTrig(q));

    % Plot
    figure('Name', sprintf('Trigonometric Fourier Series %d Terms', M));

    % Approximation
    subplot(3,1,1);
    plot(t,x,'k', 'LineWidth',2);
    hold on;
    plot(t,x_trig, 'b--', 'LineWidth',1.5);
    grid on;
    xlabel('Time t');
    ylabel('Amplitude');
    title(sprintf('Trigonometric FS - %d Terms, Approximation = %.6f%%', M, PctTrig(q)));
    legend('Original x(t)', 'Fourier approximation');
    xlim([0 T]);
    ylim([-0.1 1.2]);

    % a_n coefficients
    subplot(3,1,2);
    stem([0 n], [a0 an], 'filled');
    grid on;
    xlabel('n');
    ylabel('a_n');
    title('Cosine Coefficients a_n (including a_0)');
```

```
    % bn coefficients
    subplot(3,1,3);
    stem([0 n], [0 bn], 'filled');
    grid on;
    xlabel('n');
    ylabel('b_n');
    title('Sine Coefficients b_n');
end

%% PART (c): COSINE WITH PHASE FOURIER SERIES
fprintf('\n');
fprintf(' PART (c): COSINE WITH PHASE FOURIER SERIES\n');

for q = 1:length(termCounts)
    M = termCounts(q);
    K = Klist(q);
    n = 1:K;

    % Trigonometric coefficients
    a0 = 4/3;
    an = 3*(cos(2*pi*n/3)-1) ./ (pi^2*n.^2);
    bn = zeros(size(n));

    % Convert to magnitude-phase form
    An = sqrt(an.^2 + bn.^2);
    phi_n = atan2(-bn,an);

    % Phase is undefined if amplitude = 0
    phi_n(An<1e-12) = 0;

    % Construct approximation
    theta = (n(:)*w0)*t;

    % Add one phase to every corresponding harmonic row
    thetaPhase = bsxfun(@plus,theta,phi_n(:));
    x_phase = a0/2 + An*cos(thetaPhase);
    phaseApprox{q} = x_phase;

    % Approximation percentage
    errorEnergy = trapz(t, (x - x_phase).^2);
    PctPhase(q) = 100*(1 - errorEnergy/signalEnergy);

    fprintf('%2d terms: %.6f %% approximation\n', M, PctPhase(q));

    % Plot
    figure('Name', sprintf('Cosine Phase Fourier Series %d Terms', M));

    % Approximation
```

```
    subplot(3,1,1);
    plot(t,x, 'k', 'LineWidth', 2);
    hold on;
    plot(t,x_phase, 'm--', 'LineWidth',1.5);
    grid on;
    xlabel('Time t');
    ylabel('Amplitude');
    title(sprintf('Cosine-Phase FS - %d Terms, Approximation = %.6f%%', M, PctPhase(q)));
    legend('Original x(t)', 'Fourier approximation');
    xlim([0 T]);
    ylim([-0.1 1.2]);

    % Amplitude coefficients
    subplot(3,1,2);
    stem(n,An, 'filled');
    grid on;
    xlabel('n');
    ylabel('A_n');
    title('Amplitude Coefficients A_n');

    % Phase coefficients
    subplot(3,1,3);
    stem(n,phi_n, 'filled');
    grid on;
    xlabel('n');
    ylabel('\phi_n (rad)');
    title('Phase Coefficients \phi_n');
end

%% PART (d) AND PART (e): SUMMARY OF APPROXIMATION PERCENTAGES
fprintf('\n\n');
fprintf(' APPROXIMATION SUMMARY\n');
fprintf(' Terms    Trigonometric FS    Cosine-Phase FS    Complex FS\n');

for q = 1:length(termCounts)
    fprintf('%5d      %10.6f %%      %10.6f %%      %10.6f %%\n', termCounts(q),
PctTrig(q), PctPhase(q), PctComplex(q));
end

% Plot 3, 11 and 51 term approximations together
figure('Name', 'Comparison of 3, 11 and 51 Terms');
plot(t,x, 'k', 'LineWidth',2);
hold on;
plot(t,complexApprox{1}, 'r--', 'LineWidth',1.3);
plot(t,complexApprox{2},'b-.', 'LineWidth',1.3);
plot(t,complexApprox{3},'m:', 'LineWidth',1.8);
grid on;
xlabel('Time t');
ylabel('x(t)');
```

```
title('Comparison of 3, 11 and 51-Term Fourier Approximations');
legend('Original', '3 terms', '11 terms', '51 terms');
xlim([0 T]);
ylim([-0.1 1.2]);

%% PART (f): LINE SPECTRA
K = 25;
k = -K:K;
Ck = zeros(size(k));
Ck(k==0) = 2/3;
nz = (k~=0);
Ck(nz) = 3*(cos(2*pi*k(nz)/3)-1) ./ (2*pi^2*k(nz).^2);

% Frequency location of every spectral line
omega = k*w0;
% Magnitude
Cmag = abs(Ck);
% Phase
Cphase = angle(Ck);

% Remove meaningless phase values for zero coefficients
Cphase(Cmag < 1e-12) = 0;

figure('Name', 'Complex Fourier Series Line Spectrum');
% Magnitude line spectrum
subplot(2,1,1);
stem(omega, Cmag, 'filled');
grid on;
xlabel('\omega = k\omega_0 (rad/s)');
ylabel('|C_k|');
title('Magnitude Line Spectrum');

% Phase line spectrum
subplot(2,1,2);
stem(omega, Cphase, 'filled');
grid on;
xlabel('\omega = k\omega_0 (rad/s)');
ylabel('\angle C_k (rad)');
title('Phase Line Spectrum');

%% PART (g): THREE "GOOD" APPROXIMATIONS OVER FIVE PERIODS
t5 = linspace(0,5*T,15001);
% Convert t5 into one-period variable
tau = mod(t5,T);

% Original periodic signal over five periods
x5 = zeros(size(tau));
index1 = (tau >= 0) & (tau < 1);
x5(index1) = tau(index1);
```

```
index2 = (tau >= 1) & (tau < 2);
x5(index2) = 1;
index3 = (tau >= 2) & (tau <= 3);
x5(index3) = 3 - tau(index3);

% 51-term Complex Exponential FS
K = 25;
k = -K:K;
Ck = zeros(size(k));
Ck(k==0) = 2/3;
nz = (k~=0);
Ck(nz) = 3*(cos(2*pi*k(nz)/3)-1) ./ (2*pi^2*k(nz).^2);
E5 = exp(1i*(k(:)*w0)*t5);
xComplex5 = real(Ck*E5);

% 51-term Trigonometric FS
n = 1:K;
a0 = 4/3;
an = 3*(cos(2*pi*n/3)-1) ./ (pi^2*n.^2);
bn = zeros(size(n));
theta5 = (n(:)*w0)*t5;
xTrig5 = a0/2 + an*cos(theta5) + bn*sin(theta5);

% 51-term Cosine-with-Phase FS
An = sqrt(an.^2 + bn.^2);
phi_n = atan2(-bn,an);
phi_n(An < 1e-12) = 0;
thetaPhase5 = bsxfun(@plus, theta5, phi_n(:));
xPhase5 = a0/2 + An*cos(thetaPhase5);

% Plot all three good approximations over five periods
figure('Name', 'Three Good Approximations Over Five Periods');

% Complex exponential
subplot(3,1,1);
plot(t5,x5, 'k', 'LineWidth', 1.5);
hold on;
plot(t5, xComplex5, 'r--', 'LineWidth', 1.2);
grid on;
xlabel('Time t');
ylabel('x(t)');
title('51-Term Complex Exponential Fourier Series');
legend('Original', 'Complex FS');
xlim([0 5*T]);
ylim([-0.1 1.2]);

% Trigonometric
subplot(3,1,2);
plot(t5,x5, 'k', 'LineWidth', 1.5);
```

```
hold on;
plot(t5,xTrig5, 'b--', 'LineWidth',1.2);
grid on;
xlabel('Time t');
ylabel('x(t)');
title('51-Term Trigonometric Fourier Series');
legend('Original', 'Trigonometric FS');
xlim([0 5*T]);
ylim([-0.1 1.2]);

% Cosine with phase
subplot(3,1,3);
plot(t5,x5, 'k', 'LineWidth',1.5);
hold on;
plot(t5, xPhase5, 'm--', 'LineWidth',1.2);
grid on;
xlabel('Time t');
ylabel('x(t)');
title('51-Term Cosine-with-Phase Fourier Series');
legend('Original', 'Cosine-Phase FS');
xlim([0 5*T]);
ylim([-0.1 1.2]);

% FINAL CHECK
% The three representations must produce practically the same 51-term approximation.
difference1 = max(abs(xComplex5 - xTrig5));
difference2 = max(abs(xComplex5 - xPhase5));

fprintf('\n');
fprintf(' FINAL NUMERICAL CHECK\n');
fprintf('Max difference: Complex vs Trigonometric = %.3e\n', difference1);
fprintf('Max difference: Complex vs Cosine-Phase = %.3e\n', difference2);
```

Simulation Output Plot & Command Window Results:

Command Window

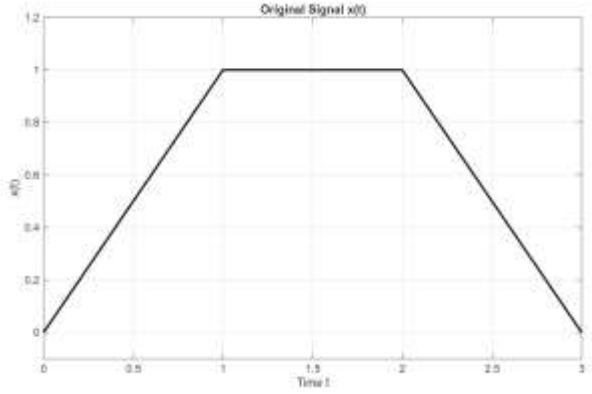
```
 PART (a): COMPLEX EXPONENTIAL FOURIER SERIES
  3 terms: 98.709751 % approximation
 11 terms: 99.982132 % approximation
 51 terms: 99.999749 % approximation

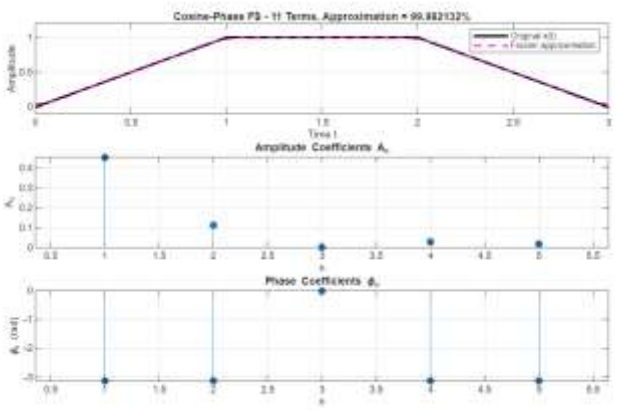
 PART (b): TRIGONOMETRIC FOURIER SERIES
  3 terms: 98.709751 % approximation
 11 terms: 99.982132 % approximation
 51 terms: 99.999749 % approximation

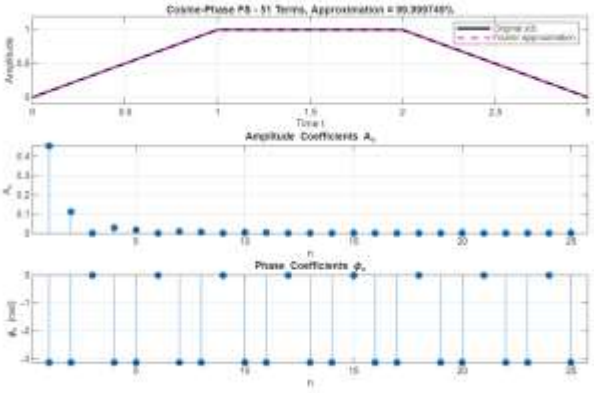
 PART (c): COSINE WITH PHASE FOURIER SERIES
  3 terms: 98.709751 % approximation
 11 terms: 99.982132 % approximation
 51 terms: 99.999749 % approximation

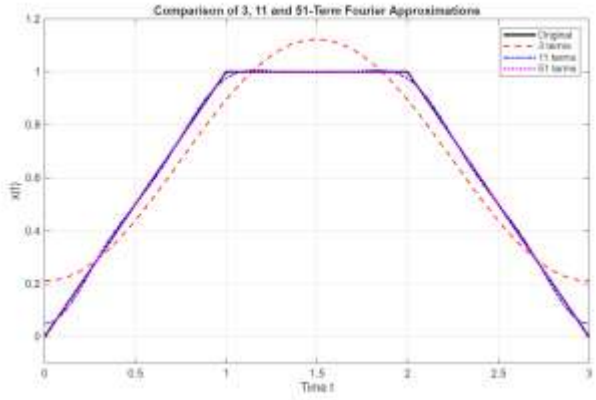

 APPROXIMATION SUMMARY
 Terms    Trigonometric FS    Cosine-Phase FS    Complex FS
    3       98.709751 %       98.709751 %       98.709751 %
   11       99.982132 %       99.982132 %       99.982132 %
   51       99.999749 %       99.999749 %       99.999749 %

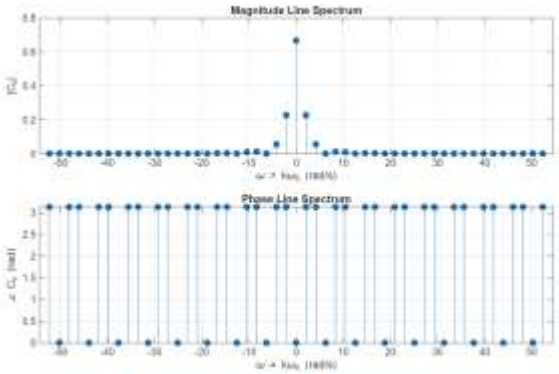
 FINAL NUMERICAL CHECK
Max difference: Complex vs Trigonometric = 6.661e-16
Max difference: Complex vs Cosine-Phase = 1.110e-15
>>
```

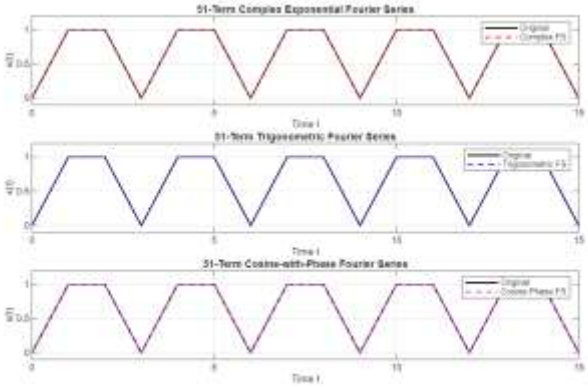


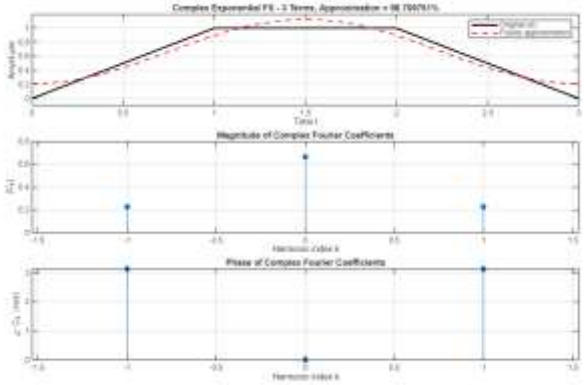


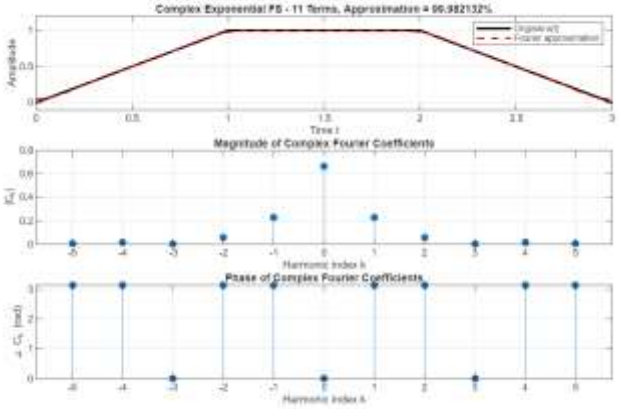


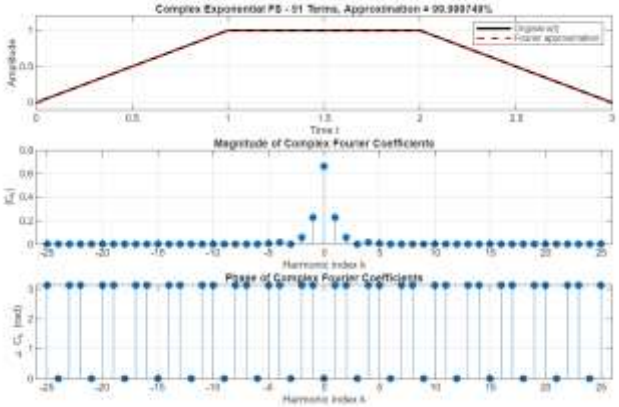

Comparison of 3, 11 and 51-Term Fourier Approximations
Original
3 terms
11 terms
51 terms
x(t)
Time t

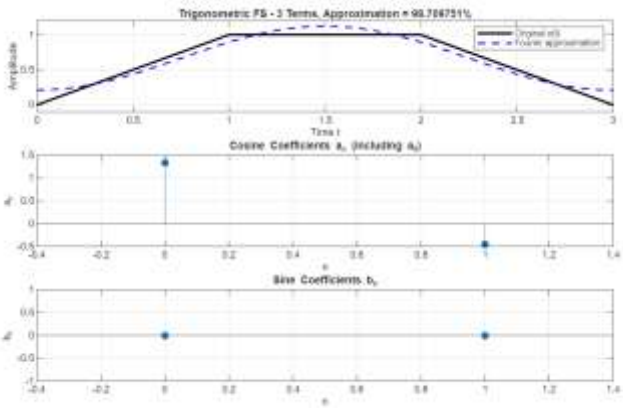

Magnitude Line Spectrum
|Cₖ|
Phase Line Spectrum
∠ Cₖ (rad)

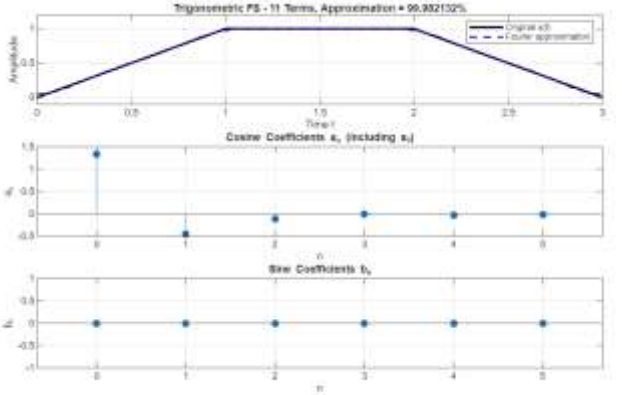

51-Term Complex Exponential Fourier Series
51-Term Trigonometric Fourier Series
51-Term Cosine-with-Phase Fourier Series
Time t

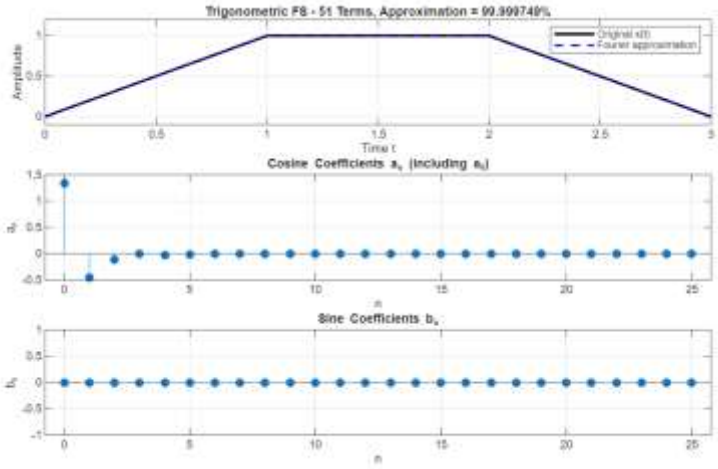

Magnitude of Complex Fourier Coefficients
Phase of Complex Fourier Coefficients
Harmonic index k

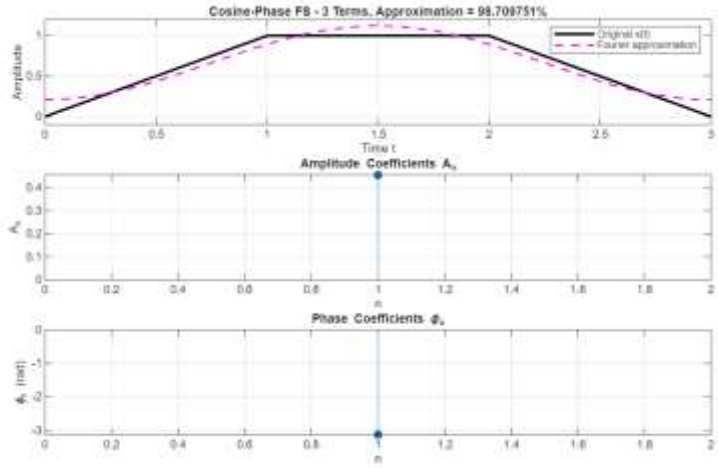

Complex Exponential FS - 11 Terms, Approximation = 99.982132%
Original x(t)
Fourier approximation
Amplitude
Time t
Magnitude of Complex Fourier Coefficients
|Cₖ|
Harmonic index k
Phase of Complex Fourier Coefficients
∠ Cₖ (rad)


Complex Exponential FS - 51 Terms, Approximation = 99.999748%
Original x(t)
Fourier approximation
Amplitude
Time t
Magnitude of Complex Fourier Coefficients
|Cₖ|
Harmonic index k
Phase of Complex Fourier Coefficients
∠ Cₖ (rad)


Trigonometric FS - 3 Terms, Approximation = 99.784751%
Original x(t)
Fourier approximation
Amplitude
Time t
Cosine Coefficients aₙ (including a₀)
Sine Coefficients bₙ
n


Trigonometric FS - 11 Terms, Approximation = 99.982132%
Original x(t)
Fourier approximation
Amplitude
Time t
Cosine Coefficients aₙ (including a₀)
Sine Coefficients bₙ
n


Trigonometric FS - 51 Terms, Approximation = 99.999749%
Original x(t)
Fourier approximation
Amplitude
Time t
Cosine Coefficients a_n (including a_0)
a_n
n
Sine Coefficients b_n
b_n
n


Cosine-Phase FS - 3 Terms, Approximation = 98.708751%
Original x(t)
Fourier approximation
Amplitude
Time t
Amplitude Coefficients A_n
A_n
n
Phase Coefficients Φ_n
Φ_n (rad)
n